Rafi Ibn Sultan

PhD Candidate in Computer Science, Wayne State University
rafi.ruet13@gmail.com | LinkedIn | GitHub | Website | Google Scholar

Research Interests

My research focuses on multimodal AI, particularly on improving reasoning and grounding in vision-language models.

Vision-Language Models · Spatial Reasoning · Multimodal Learning · Medical Image Analysis

Education

PhD Computer Science *Expected August 2027*
Advisor: Prof. Dongxiao Zhu
Wayne State University, Detroit, MI, USA

MSc Computer Science *December 2025*
Wayne State University, Detroit, MI, USA

BSc Computer Science and Engineering *November 2018*
Rajshahi University of Engineering & Technology (RUET), Bangladesh

Research Experience

Graduate Research Assistant Wayne State University *September 2023 – Present*
Trustworthy AI Lab, Department of Computer Science
Detroit, MI, USA

- Lead research in multimodal learning spanning spatial reasoning, pedestrian accessibility, and medical image segmentation.
- Contributed to the technical design and writing of the successful NIH/NEI grant proposal *VisionWay* (R61EY037504).

Clinical Research Collaboration Henry Ford Health

- Developed and validated a deep learning method for left anterior descending artery segmentation from non-contrast planning CT, in collaboration with medical physicists and clinical researchers.

Teaching and Mentoring

Graduate Teaching Assistant Wayne State University *August 2022 – August 2023*
Detroit, MI, USA

- Led Software Engineering labs, supervising 5 to 6 project teams of 4 to 5 students each per term through the full software development lifecycle, from requirements to delivery.

Lecturer Varendra University *October 2019 – August 2022*
Rajshahi, Bangladesh

- Served as instructor of record for undergraduate courses including Microprocessor and Assembly Language, Introduction to Object-Oriented Programming, and Computer Fundamentals, with full responsibility for lectures, assessments, and grading.
- Supervised more than 8 undergraduate final-year thesis and capstone projects from proposal to defense.

Graduate Research Mentoring: Mentor 2 junior PhD researchers in experimental design, implementation, evaluation, and academic writing.

Guest Teaching: Taught the large vision-language model portion of CSC 5991, *Trustworthy AI for Large Language and Vision-Language Models*, at Wayne State University (Winter 2026).

Selected Publications

First-Author Publications

[1] **RI Sultan**, H Zhu, X Zhou, C Li, P Khanduri, M Brocanelli, D Zhu. "WalkGPT: Grounded Vision-Language Conversation with Depth-Aware Segmentation for Pedestrian Navigation," *IEEE/CVF Conference on Computer Vision and Pattern Recognition* (**CVPR**), 2026. [Paper] [Code]

[2] **RI Sultan**, C Li, Y Demetriou, AI Ghanem, JP Kim, J Cunningham, H Bagher-Ebadian, D Zhu, KS Thind. "A Neighborhood Attention Transformer Network for Enhanced 3D Segmentation of the Left Anterior Descending Artery," ***Medical Physics***, 2026. [Paper] [Code]

[3] **RI Sultan**, H Zhu, C Li, D Zhu. "MedPlex: Deep Vision-Language Co-Adaptation for Clinically Grounded Medical Segmentation," *British Machine Vision Conference* (**BMVC**), 2026. [Paper] [Code]

[4] **RI Sultan**, C Li, H Zhu, P Khanduri, M Brocanelli, D Zhu. "GeoSAM: Fine-Tuning SAM with Multi-Modal Prompts for Mobility Infrastructure Segmentation," *European Conference on Artificial Intelligence* (**ECAI**), 2025. [Paper] [Code]

First-Author Manuscripts Under Review

[5] **RI Sultan**, MSA Chowdhury, S Zare Zade, C Li, P Khanduri, M Brocanelli, D Zhu. "Soft Spatial Reasoning," under review, 2026. [Paper] [Code]

[6] **RI Sultan**, X Zhou, MSA Chowdhury, C Li, P Khanduri, M Brocanelli, D Zhu. "Spatial-CORE: Confidence-Aware Grounded Spatial Reasoning in Large Vision–Language Models," under review, 2026. [Paper] [Code]

Collaborative Publications

[7] X Zhou, S Zare Zade, **RI Sultan**, A Kotov, D Zhu. "Towards Mitigating Deceptive Safety Alignment in Large Reasoning Models," *Conference on Neural Information Processing Systems* (**NeurIPS**), 2026.

[8] A Sarker, N Shah, **RI Sultan**, R Jang, D Zhu, P Khanduri. "Partition-Aware Unlearning for Removing Spurious Correlations in Large Vision-Language Models," *Conference on Neural Information Processing Systems* (**NeurIPS**), Evaluations and Datasets Track, 2026.

[9] P Khanduri, C Li, **RI Sultan**, A Sarker, Y Qiang, J Kliewer, D Zhu. "On Federated Compositional Optimization: Algorithms, Analysis, and Guarantees," *Transactions on Machine Learning Research* (**TMLR**), 2026.

[10] U Mgboh, **RI Sultan**, J Kim, K Thind, D Zhu. "FluenceFormer: Transformer-Driven Multi-Beam Fluence Map Regression for Radiotherapy Planning," *Medical Imaging with Deep Learning* (**MIDL**), 2026.

[11] C Li, **RI Sultan**, H Bagher-Ebadian, Y Qiang, K Thind, D Zhu, IJ Chetty. "Enhancing CT Image Segmentation Accuracy Through Ensemble Loss Function Optimization," ***Medical Physics***, 2025.

[12] C Li, **RI Sultan**, P Khanduri, Y Qiang, C Indrin, D Zhu. "AutoProSAM: Automated Prompting SAM for 3D Multi-Organ Segmentation," *IEEE/CVF Winter Conference on Applications of Computer Vision* (**WACV**), 2025.

[13] C Li, H Zhu, **RI Sultan**, HB Ebadian, P Khanduri, C Indrin, K Thind, D Zhu. "MulModSeg: Enhancing Unpaired Multi-Modal Medical Image Segmentation with Modality-Conditioned Text Embedding and Alternating Training," *IEEE/CVF Winter Conference on Applications of Computer Vision* (**WACV**), 2025.

[14] C Li, H Bagher-Ebadian, **RI Sultan**, M Elshaikh, B Movsas, D Zhu, IJ Chetty. "A New Architecture Combining Convolutional and Transformer-Based Networks for Automatic 3D Multi-Organ Segmentation on CT Images," ***Medical Physics***, 2023.

[15] C Li, Y Qiang, **RI Sultan**, H Bagher-Ebadian, P Khanduri, IJ Chetty, D Zhu. "FocalUNETR: A Focal Transformer for Boundary-Aware Prostate Segmentation Using CT Images," *International Conference on Medical Image Computing and Computer-Assisted Intervention* (**MICCAI**), 2023.

Invited Talks & Public Engagement

- "WalkGPT: Pixel-Grounded Navigation Guidance for Pedestrians," Voxel51 Best of CVPR 2026 Series, July 2026 [Video].
- "WalkGPT: Grounded Vision-Language Conversation with Depth-Aware Segmentation for Pedestrian Navigation," Cohere Labs Open Science Community, May 2026 [Video].
- "Multimodal AI for Mobility Infrastructure and Pedestrian Guidance," Wayne State University Departmental Seminar, 2026.
- Interviewed for "Wayne State University's AI for Mobility Project Seeks to Improve Detroit's Public Transit System," *One Detroit*, Detroit PBS, March 2024 [Video].

Awards and Honors

- **Outstanding Graduate Research Assistant Award**, Department of Computer Science, Wayne State University, 2025.

Professional Service

- **Conference Reviewer:** ICLR 2027; AAAI 2027; NeurIPS 2026; ACL Rolling Review (ARR) 2026; IJCAI-ECAI 2026; BMVC 2026; IJCNN 2025.
- **Journal Reviewer:** Pattern Recognition; Computer Vision and Image Understanding (CVIU); IEEE Transactions on Circuits and Systems for Video Technology (TCSVT); Knowledge-Based Systems; Computer Methods and Programs in Biomedicine; Computers in Biology and Medicine; Biomedical Signal Processing and Control; Results in Engineering; Medical Image Analysis.

Technical Expertise

- **Programming and Systems:** Python, PyTorch, Linux, Git, LaTeX, multi-GPU deep learning workflows.
- **Models and Methods:** Vision transformers, SAM, large vision-language models, LoRA/parameter-efficient adaptation, contrastive learning, segmentation, grounding, representation learning.
- **Tools and Libraries:** Hugging Face, MONAI, OpenCV, scikit-learn, Weights & Biases.

References

Dongxiao Zhu, PhD
Professor, Department of Computer Science
Director, Institute for AI and Data Science
Wayne State University
dzhu@wayne.edu

Marco Brocanelli, PhD
Assistant Professor
Department of Electrical and Computer Engineering
The Ohio State University
brocanelli.1@osu.edu

Kundan Thind, PhD, FCCPM
Division Head, Physics
Department of Radiation Oncology, Henry Ford Health
kthind1@hfhs.org

Prashant Khanduri, PhD
Assistant Professor
Department of Computer Science
Wayne State University
khanduri.prashant@wayne.edu